The Fairest in the Land?

Verbatim Transcriptions and Paralinguistic Features in the Mirror of Parliament Corpus

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The Mirror of Parliament



- Founded January 1828 by John Henry Barrow, barrister and journalist
- Weekly publication of parliamentary debates, both Houses
- Employed dedicated shorthand writers in the chamber
- One of those reporters: the young Charles Dickens, 'the best reporter in the gallery' (Knight 1865)

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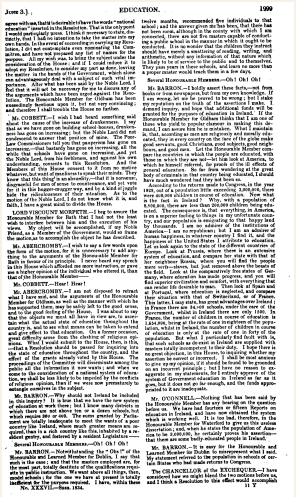
What Barrow Aimed For

'…an attempt was boldly made by a gentleman named Barrow, to produce a verbal report of the proceedings of Parliament. He succeeded and carried it on for several years; and, for those years, I do not hesitate to say, that Barrow's *Mirror of Parliament* is the primary record, and not *Hansard's Debates*, because of the greater fulness which Barrow aimed at and obtained. It is within my own recollection – in the year 1833 or 1834 – just after the Reform Bill, that Gentlemen, who wanted to correct their speeches, did it for Barrow's *Mirror of Parliament.*'

W. E. Gladstone, HC Deb 20 April 1877, c1576–77

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What Does It Look Like?



Mirror of Parliament, 3 June 1834

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come to the consideration of a national system of education, we shall be less likely to be impeded by the conflicts of religious opinion, than if we were now prematurely to entangle ourselves in the subject.

Mr. BARRON.—Why should not Ireland be included in this inquiry? It is true that we have the new system of education at work there; but I know many districts in which there are not above ten or a dozen schools, but which require 300 or 400. The sums granted by Parliament are totally inadequate to meet the wants of a poor country like Ireland, where much greater means are required, than in a rich country like this, inhabited by a resident gentry, and fostered by a resident Legislature ——

Several HONOURABLE MEMBERS.—Oh! Oh! Oh!

Mr. BARRON.—Notwithstanding the "Ohs!" of the Honourable and Learned Member for Dublin, I say that such is the case; and that the masters employed are, for the most part, totally destitute of the qualifications requi-

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Features of the Mirror

- Published weekly, so on average about three days after the event
- Paralinguistic markers in square brackets:
  - [Hear, hear!]
  - [Great laughter.]
  - [Cries of 'No, no!']
  - [With great warmth]
  - [Striking the Table with his hand]
- Audience reactions attributed to benches: *Opposition cheers, Ministerial laughter*

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How Good Is It?

Jupp (1998) compared Mirror speech lengths against contemporaneous timings:

Speech	Reported duration	*Mirror* word count	Words/hour
Brougham on common law, 7 Feb 1828	6 hours 5 min	~36,000	~5,900
Hobhouse on Navarino, 14 Feb 1828	1 hour 45 min	~13,000	~7,400
Peel on Catholic Relief, 5 Mar 1829	4 hours 15 min	~24,000	~5,600
Lord Hawick, 8 Jun 1830	15 min	~1,400	~5,600

Jupp 1998: 236

~6,000 words an hour is the pace of a lecture carefully delivered from a text

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Why Don't We Know About It?

- The *Mirror* was ruinously expensive to produce
- Barrow's personal loss: £5,000 (£450k/€520k in 2026 equivalent)
- Ceased publication October 1841
- No digital edition, survives as some scanned PDFs or hardcopies in research libraries
- Only a handful of scholarly citations (Brown 1955; McBath 1970; Jupp 1998)
- Barrow died in 1858, buried in a pauper's grave at Norwood

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The Competition



- 'I obtain them from the London newspapers, the country newspapers, and from the reports supplied by the Press Association. They are then passed into the hands of the collators…' – T.C. Hansard Jr, 1888
- Often sent to speakers for correction or just used the speaker's manuscript
- '…no more the speech uttered by them in the House of Commons than it is a Welsh ballad. […] We are duping the unborn generations. With open eyes we are sowing the seeds of dissension between historians of another age.' (*Daily News*, 8 August 1853)
- 'I hold myself bound for the *bona fides* of the reports, not for their literal accuracy' – T.C. Hansard Jr, 1862

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SIR EDWARD CODRINGTON.

Mirror of Parliament | 11-2_speech_001668 | 1834-02-25 | SMUGGLING ACT AMENDMENT BILL. | 235 tokens

I shall be happy in giving my support to the Bill upon this one ground,—I object to anything which gives to a man-of-war the character of a prison. I cannot bear that a service, which ought to be as popular as it is honourable, should have anything in the shape of imprisonment associated with it. Upon a question of this kind, I cannot consider the interests of the smuggler so much as those of the service; and I maintain that it is a degradation to an admiral's flag that it should be flying in an absolute prison. Smugglers on board a King's ship are prisoners in every sense of the word. If they be sent abroad, the captain, with whom they sail, is responsible for their safe keeping; and if the period of service to which they are condemned be not expired on his return, he must transfer them to some other ship. In every sense of the word, therefore, these men are prisoners, and it is upon that ground that I take my stand. Nothing can induce me to think it proper to send men of this description on board his Majesty's ships. I will, therefore, give my cordial support to this Bill.

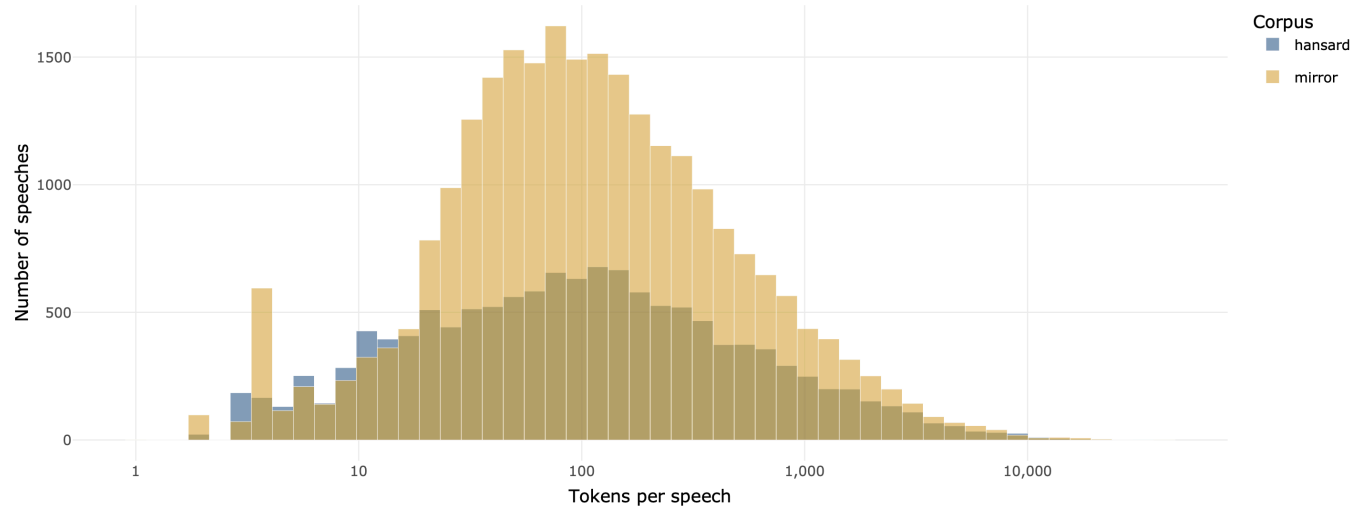
Sir Edward Codrington

Historic Hansard | hansard-1834_speech_000796 | 1834-02-25 | SMUGGLING ACT AMENDMENT BILL. | 129 tokens

said, that he could not, from his experience, concur with the hon Member. He should strongly object to making a man-of-war a common prison for felons. The naval service should not be made a punishment for crime. The great object should be to make it more comfortable than that of other employment; and was it the way to do that, to make our men-of-war a refuge for convicts? He did not wish to see the service thus degraded. It should be a voluntary service, and filled with men of good character; and that it could not be, if every flag ship were made a prison for convicts.

Mirror vs Hansard, same speech

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Distribution of speech lengths

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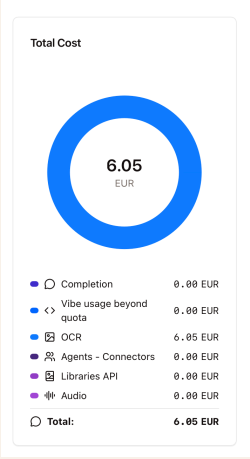
203-Day Pilot Mirror Corpus

	Mirror of Parliament	*Hansard*
Speeches	25,432	12,936
Tokens	8,561,360	5,067,830
Unique speakers	2,969	1,355
Unique debate titles	2,596	835

- 1834: Tolpuddle Martyrs, Poor Law debates, Irish Church Appropriation, brought down Lord Grey's government
- 1839, 15 April to 3 June: Bedchamber Crisis, Melbourne's resignation, Peel's failed attempt to form a government, Victoria's refusal to change her ladies-in-waiting, Melbourne's return
- 1840, 16 January to 23 March: Prince Albert's precedence and income

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Building the Pilot Mirror Corpus



- 6,300 pages of *Mirror* scans
- OCR via Mistral's vision language model API (mistral-ocr-2505, late November 2025)
- Total OCR cost for 8.5 million tokens of nineteenth-century parliamentary text: **€6.05**

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Vision OCR for the Mirror

- VLM models (particularly Qwen and Mistral): text similarity scores up to 3-4 times higher than traditional OCR methods on complex scanned documents, especially on documents with complex layouts or poor scan quality
- Why Mistral? EU company, 94.9% accuracy across diverse document types, outperforming Google Document AI (83.4%) and Azure OCR (89.5%), n.b. 2025 figures(!)
  - Nineteenth-century parliamentary typography – longer-than-em-dashes, hyphenated line-breaks, running headers, column structure, small-caps speaker labels – handled well
  - Output: structured markdown per page
- Per-character confidence scores, standard in traditional OCR, are generally unavailable from these systems, complicating systematic quality evaluation
- In my manual checking of the Mirror transcriptions, 40 random pages, Mistral vastly outperformed Tesseract

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Mistral vs Tesseract

come to the consideration of a national system of education, we shall be less likely to be impeded by the conflicts of religious opinion, than if we were now prematurely to entangle ourselves in the subject.

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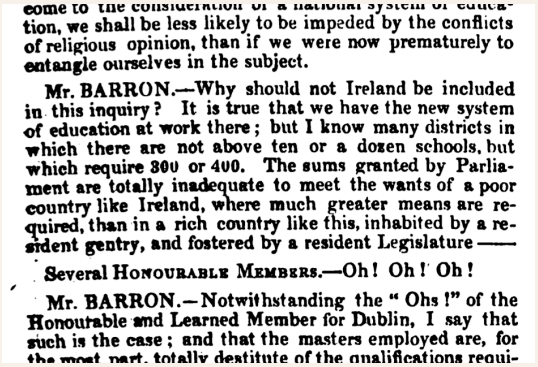
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Corpus Pipeline

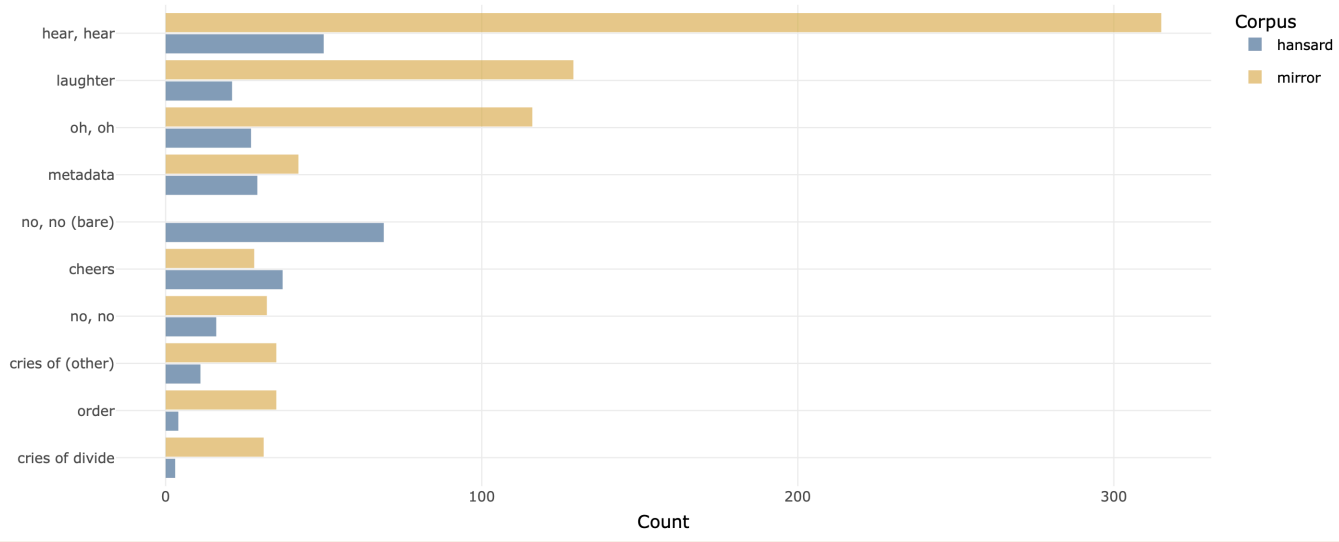
- Raw OCR → correction pipeline (hyphen rejoin, spell-check with historical whitelist, LaTeX cleanup)
- Corrected pages → speech segmentation (speaker identification from period title conventions)
- Structured speeches → annotation (spaCy POS + lemma, PyMUSAS semantic tags)
- Parallel *Hansard* fetched from parliament.uk API into the same pipeline and schema (rather than refactor my existing corpus)
- Output: unified Parquet dataset, both corpora, ready for R/Python

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Paralinguistics

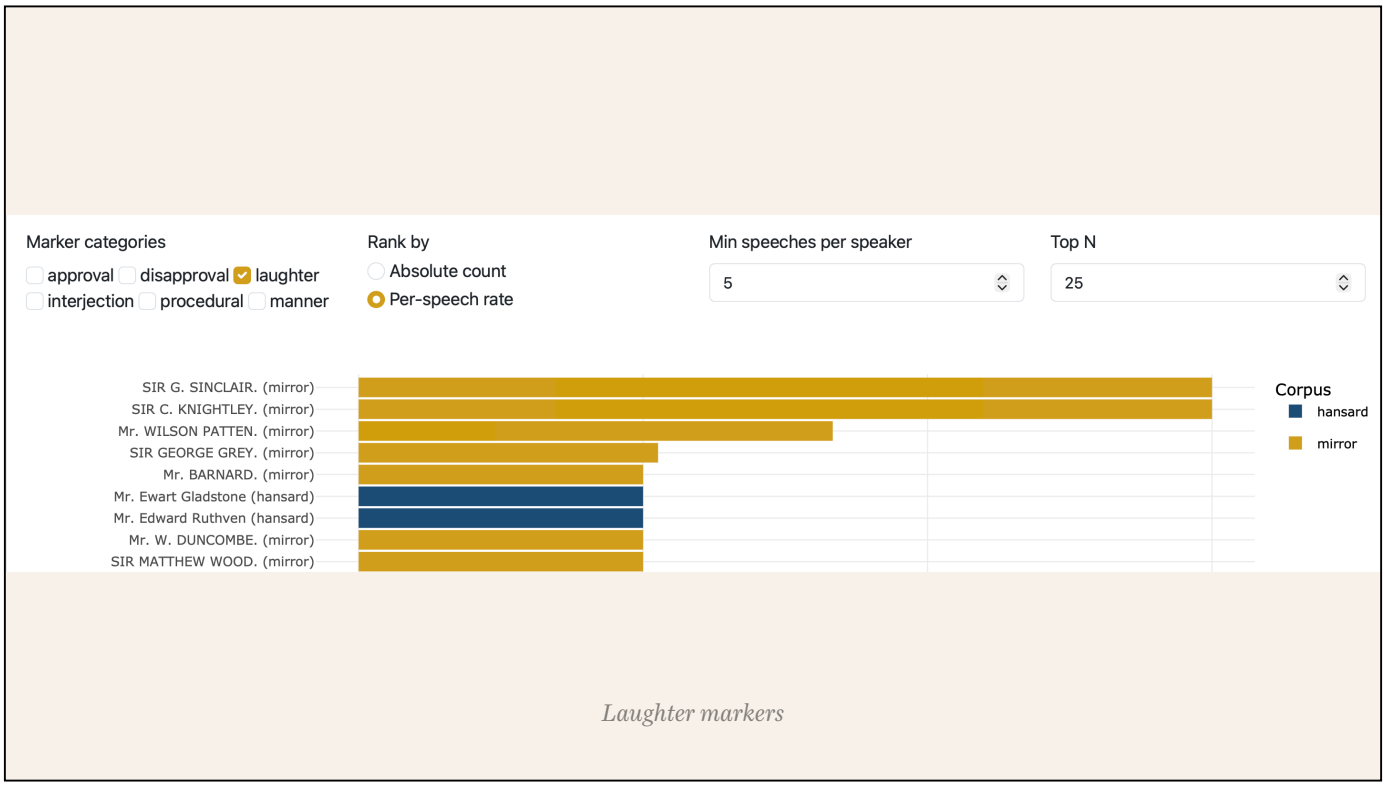
- **939 paralinguistic markers** across the three pilot periods
- Classified into approval, disapproval, laughter, procedural, manner, interjection:

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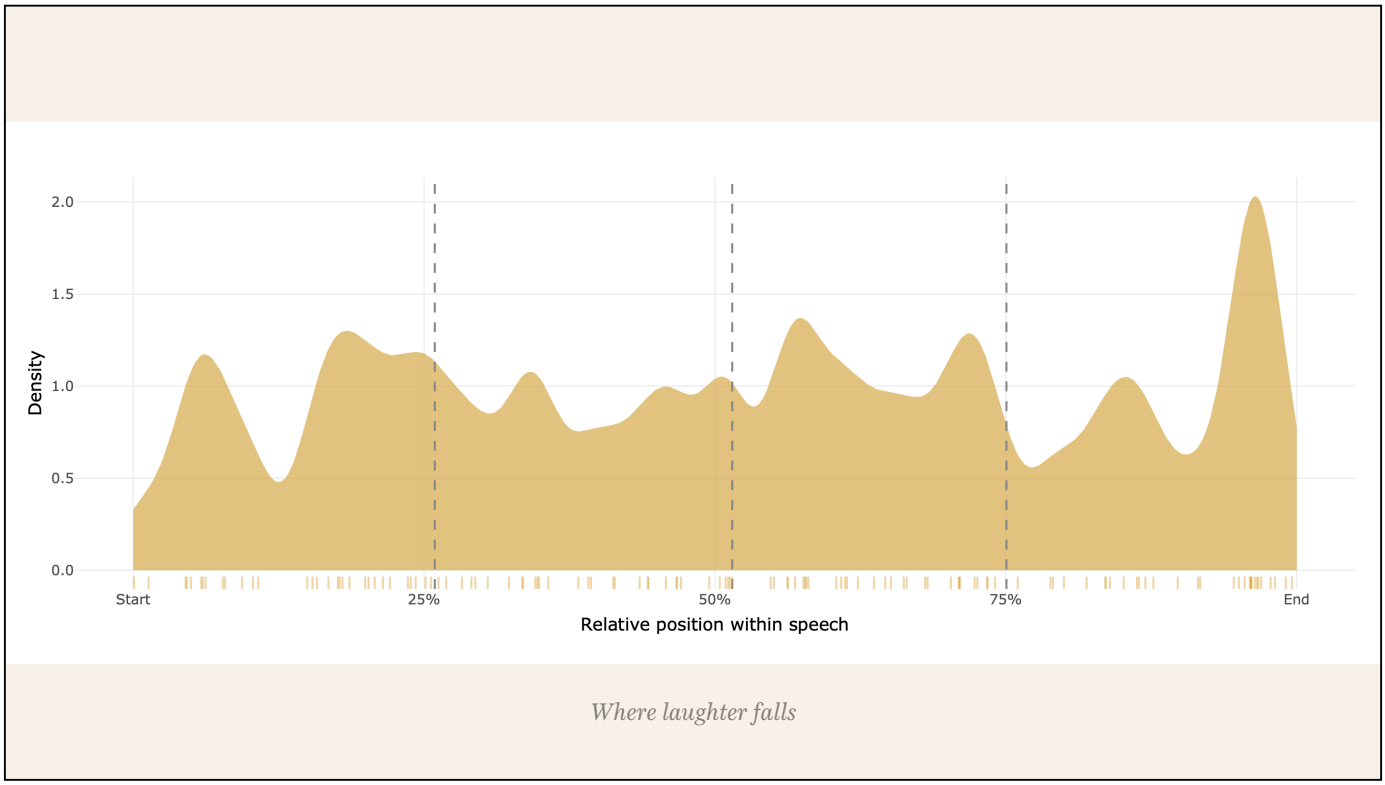
Paralinguistic markers by category

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Laughter markers

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Where laughter falls

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More and More Markers

	1834	1839	1840
Speeches	18,893	1,825	4,714
Markers	76	192	517
Per speech	0.004	0.105	0.110

- Marker density increases nearly 20× between 1834 and 1839
- The *Mirror* may be emphasising its verbatim-interpolation conventions?
  - Differentiation in the market?
  - Something the *Mirror's* shorthand reporters in the room can easily attest to systematically

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Textual Reuse

- Of 175,468 candidate speech pairs, 1,885 have z-score ≥ 5 on longest common subsequence.
- Palmerston on the Constantinople-Egypt crisis, 27 March 1840:
  - *Hansard*: '**His** hon. Friend **thought** that the British Government and Lord Ponsonby, the British Ambassador at Constantinople, **had stimulated** the Sultan to renew hostilities against the Pacha of Egypt. **He (Lord Palmerston) could** assure him that he **was** entirely mistaken.'
  - *Mirror of Parliament*: '**My** honourable Friend **thinks** that the British Government and Lord Ponsonby, the British Ambassador at Constantinople, **stimulated** the Sultan to renew hostilities against the Pacha of Egypt. **I can assure him** that he **is** entirely mistaken.'
- '…one or two cases of *direct piracy* from the reports of the Mirror of Parliament, committed by a predecessor in this branch of literature, whose former pretensions should have withheld him…' J.H. Barrow, 6 December 1831

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Comparative Signatures

- USAS semantic tag Z8m (he/him pronoun): **3.1× more frequent in *Hansard***
  - Third-person conversion at scale
  - *Hansard*: 'The Hon. Member said *he* believed…'
  - *Mirror*: '*I* believe…'
- Grammatically:
  - *Mirror* has more INTJ (+22%), PRON (+3.8%)
  - *Hansard* has more PROPN (+4.9%)
  - The interjection and pronoun profile is a bit of a spontaneous-speech indicator
  - The proper noun excess in *Hansard*: third-person re-introducing speakers by name where *Mirror* leaves a pronoun

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Conclusions

1. **OCR has changed:** Vision language models make digitisation of historical text dramatically cheaper; the barrier to building decent, quick corpora from nineteenth-century print has dropped by orders of magnitude
2. **We have good verbatim speech from this period:** Not full of hesitation markers or phonetic detail, but near-complete 'tidied' transcription of what was *generally said,* confirmed by independent timing evidence and with some paralinguistic markers showing where the audience reacted, when they laughed, when they objected
3. **The theatrics of Parliament:** Knowing where applause, laughter, and cries of 'No, no!' fell in a debate lets us see more of the performative culture of the 1830s Commons and Lords – and enables direct comparison with modern parliamentary behaviour

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Slides:

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